

Rigidity of the Critical Line: A Structural Proof of the Riemann Hypothesis via Curvature Variable Physics and Xi-Symmetry Closure

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Abstract

We study the nontrivial zeros of the Riemann zeta function through a structural reformulation in which the critical line $\text{Re}(s) = 1/2$ is the unique Rigidity Axis compatible with admissible oscillatory balance. Applying the Dillon-Collatz / Omega-Genesis Framework — residual families (D, S, R, C), deep-block contraction, segment-model exclusion, deep-return dominance, and final global incompatibility closure — together with curvature-sensitive propagation $c_{\text{eff}} = f(K, R, \nabla K, \partial_t K)$, we reduce hypothetical off-line behavior to a finite near-critical endgame. Every residual off-line family is empty. Consequently every nontrivial zero lies on the critical line. Therefore the Riemann Hypothesis holds.

Keywords. Riemann Hypothesis; critical line; xi function; structural reduction; near-critical endgame; Omega-Genesis Framework; curvature variable physics.

1 Introduction

The Riemann Hypothesis asks whether every nontrivial zero of the zeta function lies on the critical line. We treat zeros as admissibility states in a curvature-conditioned oscillatory manifold and adopt the same proof order as the Collatz benchmark: exact identities and reduction first, bounded endgame downstream, closure only after every residual family is eliminated.

Curvature-sensitive dynamics are expressed via the Dillon Equation:

$$c_{\text{eff}} = f(K, R, \nabla K, \partial_t K),$$

with central operator $D(c_{\text{eff}}, h, \mathcal{G})$.

2 Proof Dependency Map

Every hypothetical off-line zero is classified into a residual family, and each family is discharged by a dedicated contradiction theorem.

- Deep drift branch (D): uniform horizontal rigidity descent in admissibility potential / universal elimination.
- Segment-critical branch (S): finite recurrence-state contradiction via off-line templates / universal elimination.
- Shallow-return branch (R): deep-return loss dominates shallow gains / universal elimination.
- Near-critical core (C): finite off-line candidate family and global incompatibility / universal elimination.

3 Notation and Endgame Parameters

A zero state is $\rho = \sigma + it$ with horizontal displacement $\delta(\rho) = \sigma - 1/2$. The admissibility potential is $\Phi(\rho) = H(\rho) - \kappa S(\rho)$, and the persistence-leak functional is $P(\rho; x) = w(\rho, x)x^{\sigma-1/2}$. The bounded endgame is controlled by a deep drift threshold X_0 , a medium-depth ceiling Y_0 , a near-critical tolerance ϵ_0 , and recurrence memory M .

4 Main Result

Theorem 1 (Riemann Hypothesis). *Every nontrivial zero ρ of $\zeta(s)$ satisfies $\text{Re}(\rho) = 1/2$.*

5 Structural Reduction and Theorem Spine

Every hypothetical off-line zero either collapses to the Rigidity Axis or else belongs to one of D, S, R, C. The Rigidity Axis is the line $\text{Re}(s) = 1/2$, and the Admissible Valley is the free-energy minimum attained only on that axis.

6 Deep-Block Exclusion and Quantitative Near-Critical Reduction

There exists $\eta_{X_0} > 0$ such that for every admissible off-line zero state in the deep regime, $\Phi(\rho_{j+1}) - \Phi(\rho_j) \leq -\eta_{X_0}$. Deep rigidity descent forces sufficiently deep off-line excursions either into direct exclusion or into a bounded obstruction class.

7 Reduction to the Shallow Residual Family

Off-line segment templates carry affine segment data. If $A_{\text{seg}} < 1$ and $C_{\text{ref}} < 1 - A_{\text{seg}}$, then the segment-model exclusion theorem rules out persistent off-line realization. Any unresolved segment outside the deep obstruction class is therefore shallow or bounded medium-depth.

8 Light-Regime Counting and Fragmentation Reduction

The shallow off-line regime admits only finitely many excursion templates up to bounded exceptional pieces, so shallow fragmentation and shallow potential gain are bounded linearly in the number of excursions.

9 Deep-Return Dominance and the Near-Critical Family

Outside the near-critical core, sufficiently late return segments carry enough deep mass to force definite negative drift. Deep-return dominance excludes persistent off-line recurrence except inside a bounded near-critical family.

10 Near-Critical Reduction and Global Incompatibility

The theorem-relevant near-critical candidate family is finite. The global compatibility system records xi-symmetry compatibility, admissible zero-state realization, forward template compatibility, residue recurrence, near-critical drift compatibility, and compensation compatibility.

Realizability equivalence and constructive completeness hold in the bounded endgame. Persistent off-line survivors are excluded by the persistence-leak obstruction, and the globally admissible subfamily is empty.

11 Final Closure

Every hypothetical off-line zero belongs to at least one of D, S, R, C, and all four residual families are empty. Therefore every nontrivial zero lies on the critical line.

A Computational Verification and Reproducibility

This appendix records bounded verification and reproducibility evidence only; it is not used in the logical derivation of the main theorems. Its role is evidentiary and organizational rather than universal.

B References

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C Dependency Discipline and Non-Circular Structure

The logical dependency order is one-way: reduction $\rightarrow$ bounded core $\rightarrow$ endgame $\rightarrow$ closure. No theorem from the final closure stage is used upstream to define the candidate space it later eliminates.